



Assignment 3

Subnetting and Implementation in eNSP

Course: HUAWEI DATACOM

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1) Introduction

This report documents the design and implementation of an IP addressing scheme for a small network using the mother subnet 192.168.1.0/24. The network is divided into four subnets, each connected to a router interface and hosting two PCs. The objective is to configure the router, assign IP addresses, and verify connectivity through ping tests.

2) Network Design

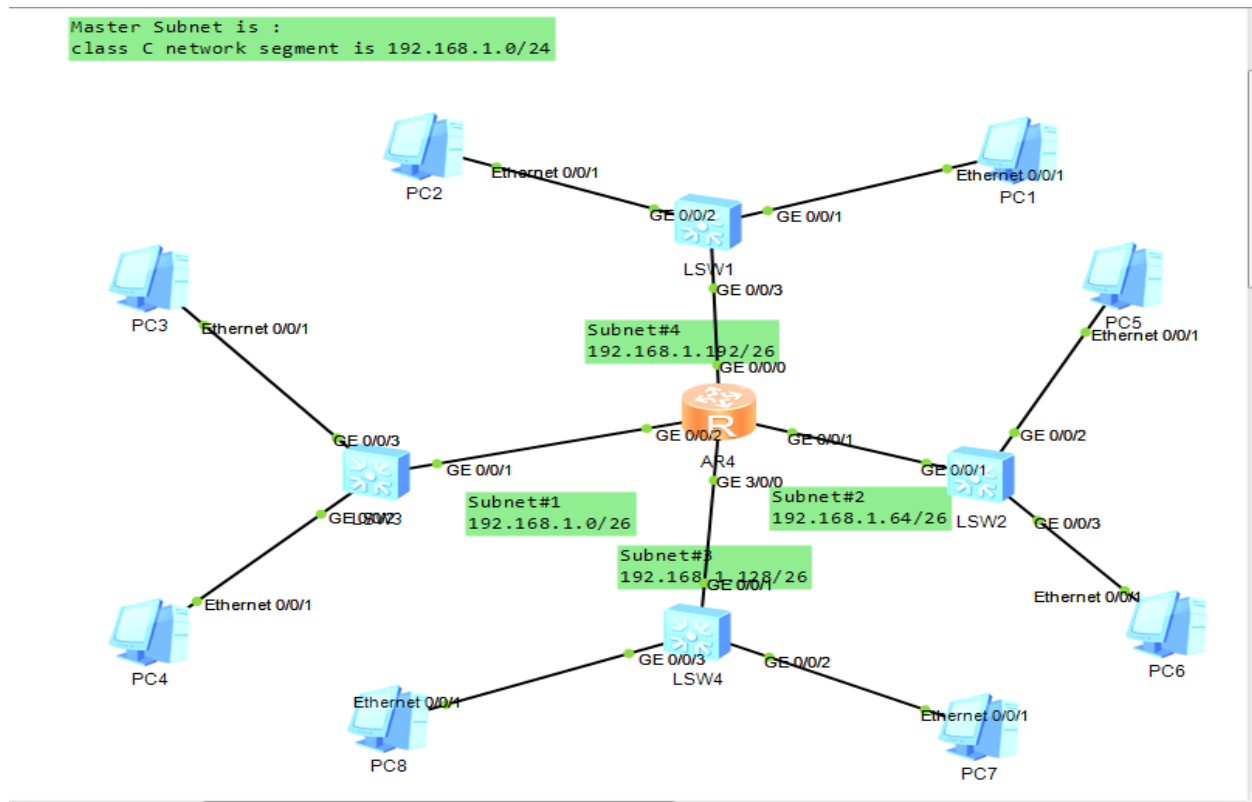
➤ Subnetting

The given Class C network 192.168.1.0/24 was divided into four equal subnets using /26 masks (255.255.255.192). Each subnet provides 62 usable host addresses.

Subnet	Network Address	Subnet Mask	Gateway	Host IPs Assigned
Subnet #1	192.168.1.0/26	255.255.255.192	192.168.1.1	PC3: 192.168.1.2, PC4: 192.168.1.3
Subnet #2	192.168.1.64/26	255.255.255.192	192.168.1.65	PC5: 192.168.1.66, PC6: 192.168.1.67
Subnet #3	192.168.1.128/26	255.255.255.192	192.168.1.129	PC7: 192.168.1.130, PC8: 192.168.1.131
Subnet #4	192.168.1.192/26	255.255.255.192	192.168.1.193	PC1: 192.168.1.194, PC2: 192.168.1.195

➤ Topology Diagram

The topology consists of a router (R1) with four GigabitEthernet interfaces, each connected to a switch with two PCs.



3) Configuration

➤ Router Configuration (R1)

Huawei VRP commands applied on R1:

```
AR4
The device is running!

<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int g0/0/2
[R1-GigabitEthernet0/0/2]ip add 192.168.1.1 255.255.255.192
[R1-GigabitEthernet0/0/2]
Sep 29 2025 03:04:23-08:00 R1 %01IFNET/4/LINK_STATE(1)[0]:The line protocol IP
on the interface GigabitEthernet0/0/2 has entered the UP state.
[R1-GigabitEthernet0/0/2]undo shutdown
Info: Interface GigabitEthernet0/0/2 is not shutdown.
[R1-GigabitEthernet0/0/2]quit
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip add 192.168.1.65 255.255.255.192
[R1-GigabitEthernet0/0/1]
Sep 29 2025 03:05:19-08:00 R1 %01IFNET/4/LINK_STATE(1)[1]:The line protocol IP
on the interface GigabitEthernet0/0/1 has entered the UP state.
[R1-GigabitEthernet0/0/1]undo shutdown
Info: Interface GigabitEthernet0/0/1 is not shutdown.
[R1-GigabitEthernet0/0/1]exit
      ^
Error: Unrecognized command found at '^' position.
[R1-GigabitEthernet0/0/1]quit
[R1]int g3/0/0
[R1-GigabitEthernet3/0/0]ip add 192.168.1.129 255.255.255.192
Sep 29 2025 03:06:24-08:00 R1 %01IFNET/4/LINK_STATE(1)[2]:The line protocol IP
on the interface GigabitEthernet3/0/0 has entered the UP state.
[R1-GigabitEthernet3/0/0]undo shutdown
Info: Interface GigabitEthernet3/0/0 is not shutdown.
[R1-GigabitEthernet3/0/0]quit
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ip add 192.168.1.193 255.255.255.192
Sep 29 2025 03:07:02-08:00 R1 %01IFNET/4/LINK_STATE(1)[3]:The line protocol IP
on the interface GigabitEthernet0/0/0 has entered the UP state.
[R1-GigabitEthernet0/0/0]undo shutdown
Info: Interface GigabitEthernet0/0/0 is not shutdown.
[R1-GigabitEthernet0/0/0]quit
[R1]save
      ^
Error: Unrecognized command found at '^' position.
[R1]quit
<R1>save
  The current configuration will be written to the device.
  Are you sure to continue? (y/n)[n]:y
  It will take several minutes to save configuration file, please wait.....
  .....
  Configuration file had been saved successfully
  Note: The configuration file will take effect after being activated
<R1>
```

➤ PC Configuration

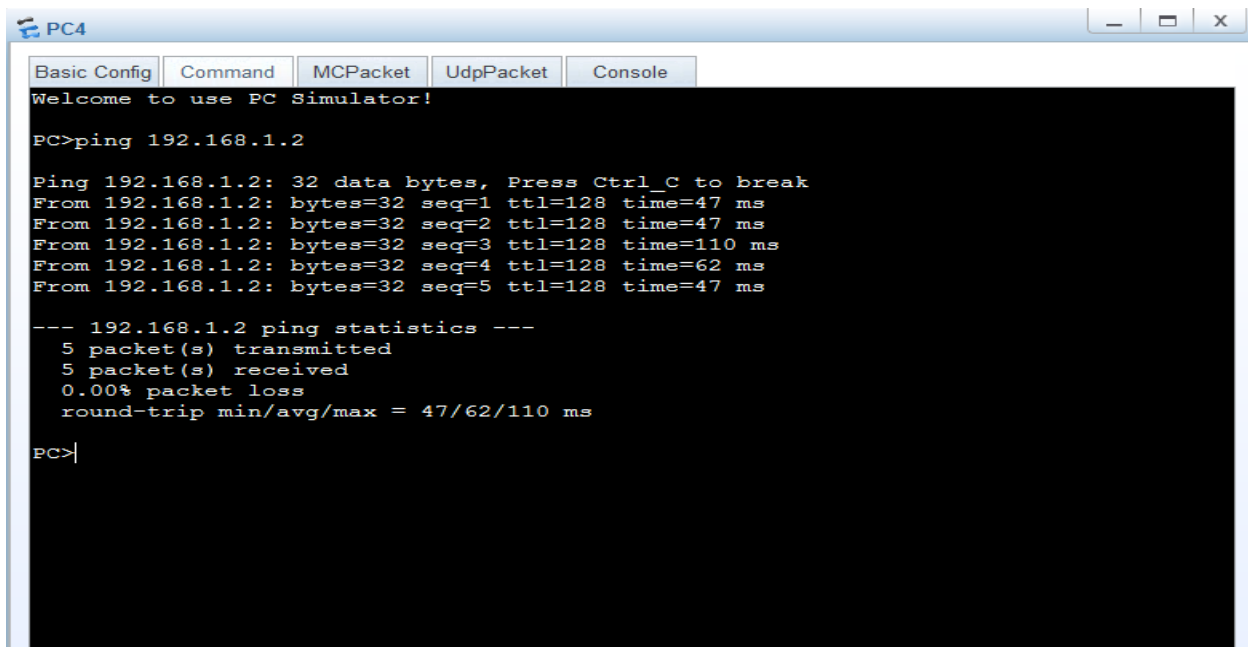
Each PC was assigned an IP address, subnet mask, and default gateway according to its subnet (see Table).

Configuration was done through the eNSP PC IP configuration tool.

4) Ping Tests

- PC4 → PC3 (same subnet)

Result: Successful (local communication works).



The screenshot shows the 'PC4' configuration window with the 'Console' tab selected. The console output displays the results of a ping command from PC4 to 192.168.1.2. The output shows five successful ping attempts with varying response times (47, 47, 110, 62, 47 ms). The statistics section indicates that all five packets were transmitted and received, resulting in 0.00% packet loss.

```
PC4
Basic Config Command MCPacket UdpPacket Console
Welcome to use PC Simulator!
PC>ping 192.168.1.2

Ping 192.168.1.2: 32 data bytes, Press Ctrl_C to break
From 192.168.1.2: bytes=32 seq=1 ttl=128 time=47 ms
From 192.168.1.2: bytes=32 seq=2 ttl=128 time=47 ms
From 192.168.1.2: bytes=32 seq=3 ttl=128 time=110 ms
From 192.168.1.2: bytes=32 seq=4 ttl=128 time=62 ms
From 192.168.1.2: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.1.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/62/110 ms

PC>|
```

- **PC1 → PC4 (different subnets via router)**

Result: Successful (inter-subnet routing works).

```
PC>ping 192.168.1.194

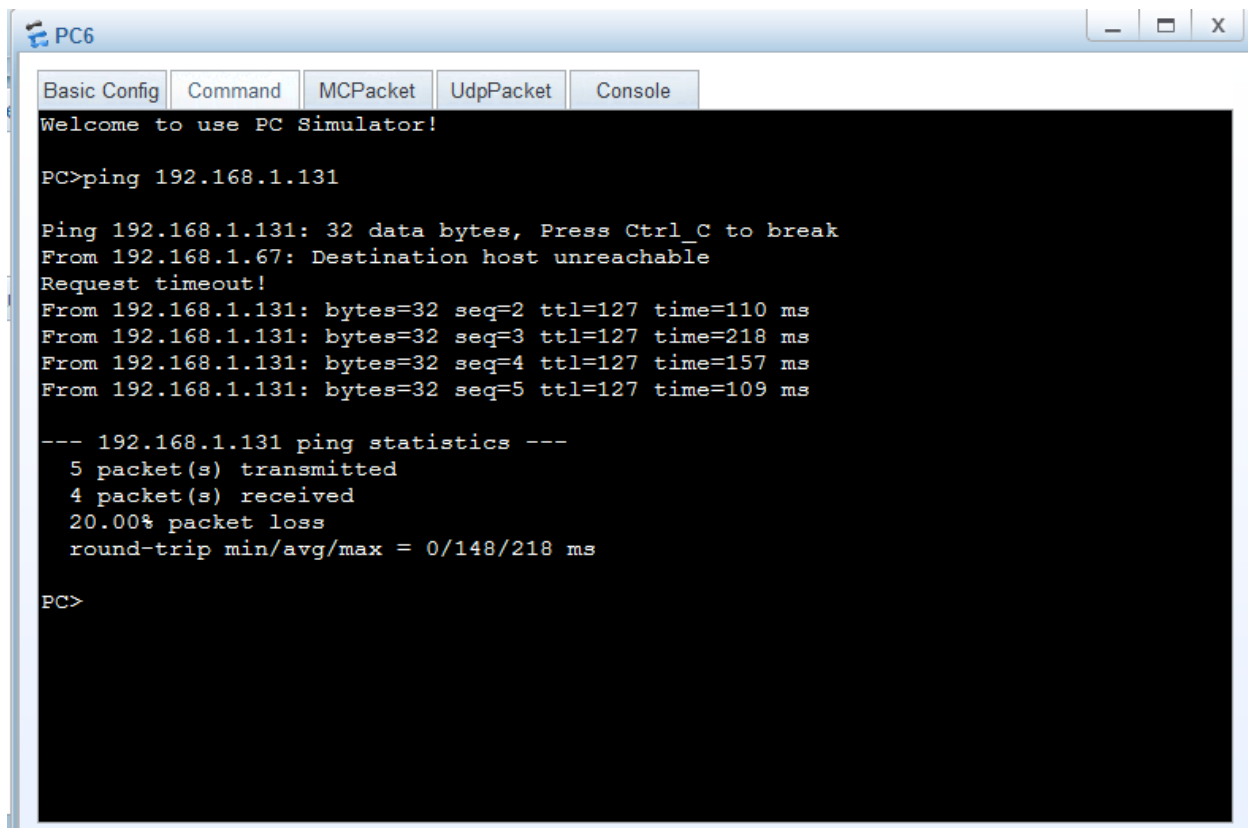
Ping 192.168.1.194: 32 data bytes, Press Ctrl_C to break
From 192.168.1.3: Destination host unreachable
Request timeout!
From 192.168.1.194: bytes=32 seq=2 ttl=127 time=515 ms
From 192.168.1.194: bytes=32 seq=3 ttl=127 time=172 ms
From 192.168.1.194: bytes=32 seq=4 ttl=127 time=234 ms
From 192.168.1.194: bytes=32 seq=5 ttl=127 time=141 ms

--- 192.168.1.194 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/265/515 ms

PC>|
```

- **PC6 → PC8 (different subnets via router)**

Result: Successful.



The screenshot shows a window titled "PC6" with a tabbed interface. The "Console" tab is active, displaying the following text:

```
Welcome to use PC Simulator!

PC>ping 192.168.1.131

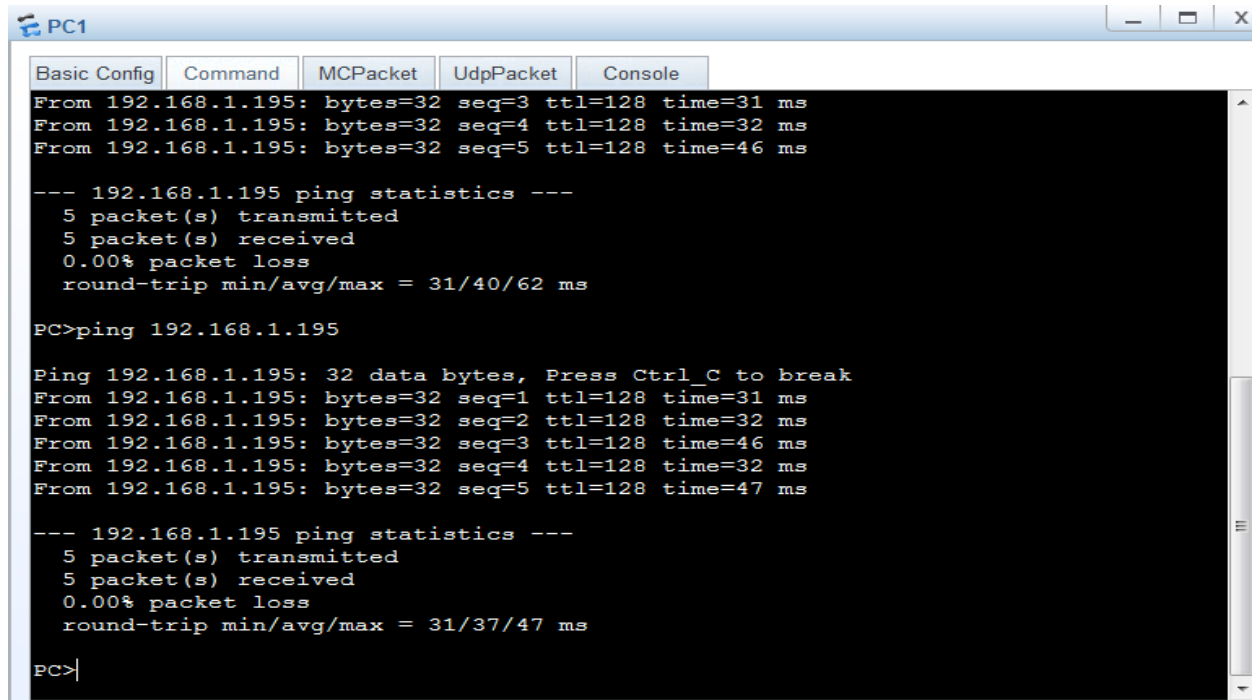
Ping 192.168.1.131: 32 data bytes, Press Ctrl_C to break
From 192.168.1.67: Destination host unreachable
Request timeout!
From 192.168.1.131: bytes=32 seq=2 ttl=127 time=110 ms
From 192.168.1.131: bytes=32 seq=3 ttl=127 time=218 ms
From 192.168.1.131: bytes=32 seq=4 ttl=127 time=157 ms
From 192.168.1.131: bytes=32 seq=5 ttl=127 time=109 ms

--- 192.168.1.131 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/148/218 ms

PC>
```


- PC1 → PC2 (same subnet)

Result: Successful.



The screenshot shows a terminal window titled 'PC1' with tabs for 'Basic Config', 'Command', 'MCPacket', 'UdpPacket', and 'Console'. The 'Console' tab is active, displaying the following text:

```
From 192.168.1.195: bytes=32 seq=3 ttl=128 time=31 ms
From 192.168.1.195: bytes=32 seq=4 ttl=128 time=32 ms
From 192.168.1.195: bytes=32 seq=5 ttl=128 time=46 ms

--- 192.168.1.195 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/40/62 ms

PC>ping 192.168.1.195

Ping 192.168.1.195: 32 data bytes, Press Ctrl_C to break
From 192.168.1.195: bytes=32 seq=1 ttl=128 time=31 ms
From 192.168.1.195: bytes=32 seq=2 ttl=128 time=32 ms
From 192.168.1.195: bytes=32 seq=3 ttl=128 time=46 ms
From 192.168.1.195: bytes=32 seq=4 ttl=128 time=32 ms
From 192.168.1.195: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.1.195 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/37/47 ms

PC>|
```

5) Conclusion

The network was successfully divided into four subnets using the 192.168.1.0/24 mother subnet. Router R1 was configured with four interfaces, each serving as a gateway for its subnet. PCs were assigned IP addresses and default gateways according to their subnets.

Verification through ping tests confirmed that:

- Hosts in the same subnet can communicate directly.
- Inter-subnet communication is enabled through R1.

This demonstrates the correctness of the IP addressing scheme and the successful routing configuration in eNSP.